

Advait Pramod Deshmukh

+91 88984-59240 · Bangalore · advait.deshmukh3@gmail.com · [linkedin.com/in/advaitdeshmukh](https://www.linkedin.com/in/advaitdeshmukh) · <https://github.com/advait-d>

SUMMARY

Software Engineer with **5+ years** building scalable distributed systems and reliable data pipelines in Java and Python. Deep experience with event-driven microservices, data correctness patterns, and high-throughput backend systems, including production ownership at Amazon scale. Adept at debugging complex defects, optimizing system performance, and collaborating across cross-functional teams to ship reliable, high-quality software.

PROFESSIONAL EXPERIENCE

Consultant – Capgemini | Bangalore, KA, India | Mar 2025 - Present

- Designed and delivered a **state-driven** credit activation service handling collateral validation, legal documentation association, and credit limit activation across **\$50B+ in collateral exposure** and multiple asset classes.
- Architected **asynchronous, event-driven workflows** across distributed microservices, integrating a **transactional outbox** pattern to guarantee atomic state changes and correctness-first publishing to Solace, achieving **~95% straight-through processing (STP)**.
- Contributed to a **large-scale data migration** spanning 22+ years of historical records for **20,000+ corporate banking clients**, reconstructing state via **Solace queue replay** within scheduled maintenance windows, as part of a Spring Boot to Quarkus, and PostgreSQL to Oracle stack modernization.
- Owned **API design, service contracts**, and **inter-service interaction patterns**, ensuring extensibility and maintainability across the platform's microservices architecture.

Senior Software Engineer – World Vision Infotech | Pune, MH, India | Sep 2024- Feb 2025

- Architected and deployed an **LLM-based agentic customer service system** using Python, OpenAI API, and FastAPI, with multi-turn context management and fallback handling, **automating 65%** of customer inquiries, **reducing support costs by 30%**.
- Designed **prompt engineering** strategies and robust **error-handling pipelines** to ensure reliable, production-grade AI system behavior at scale.

Technical Consultant – Perficient Inc | Redmond, WA, USA | Apr 2022 - Dec 2023

- Optimized Insurance Claims API processing using Java, Spring Boot, Oracle SQL, and JAX-RS, **increasing automated claims from 88% to 93%** across 12,000+ daily pharmacy orders.
- Engineered a research analytics web application backend handling **1,500 transactions per minute** using Spring Boot and PostgreSQL, supporting data ingestion, processing, and CRUD across pharmaceutical trial datasets.
- Proactively identified and resolved **250+ production issues** and security vulnerabilities, **maintaining 99.99% uptime**; improved regression test suite coverage by 24% using Cucumber/Gherkin.

Software Development Engineer - Amazon | Seattle, WA, USA | June 2020 - Mar 2022

- Owned full design and delivery of a browser-based bot challenge system serving millions of users, coordinating across 3 cross-functional teams from architecture through production rollout, reducing fraudulent sign-ups and sign-ins by 26%.
 - Redesigned CAPTCHA validation service with **asynchronous processing** in Java, reducing p99 latency from **650ms to 560ms (14% improvement)** at high-throughput scale across Amazon's authentication surface.
 - Built **event-driven workflows** across AWS DynamoDB, SQS, SNS, Lambda, and Redshift, **eliminating 80% of manual operational load** and improving system observability.
 - Led end-to-end deprecation of a legacy internal service, scoping impact via data analytics in Jupyter, driving alignment with 5+ dependent teams, and delivering 20% reduction in operational costs.
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SKILLS

- Languages:** Java, Python, C, JavaScript & TypeScript, HTML, CSS
 - Frameworks & APIs:** Spring Boot, Quarkus, FastAPI, Flask, Django, REST API, GraphQL, Apache Drools
 - Distributed Systems:** Apache Kafka, Solace, AWS SQS/SNS/Lambda/DynamoDB/Redshift, event-driven architecture
 - DevOps & Cloud:** Docker, Kubernetes, Jenkins, AWS (Certified Cloud Practitioner), CI/CD pipelines, Git, GitHub, BitBucket
 - Observability:** Grafana, Splunk, AWS CloudWatch (metrics & alerting), AWS CloudWatch Logs (log aggregation & analysis)
 - AI/ML:** OpenAI API, Claude API, GitHub Copilot, RAG pipelines, Prompt Engineering, TensorFlow, PyTorch
 - Databases:** MySQL, Oracle SQL Developer, PostgreSQL, AWS DynamoDB
 - Testing:** JUnit, Mockito, PowerMock, Selenium, Cucumber / Gherkin, Postman, JMeter
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EDUCATION

University at Buffalo, The State University of New York

Master of Science, Computer Science and Engineering

Buffalo, NY

August 2018 – February 2020

University of Mumbai

Bachelor of Engineering, Computer Engineering

Mumbai, India

August 2013 – May 2017